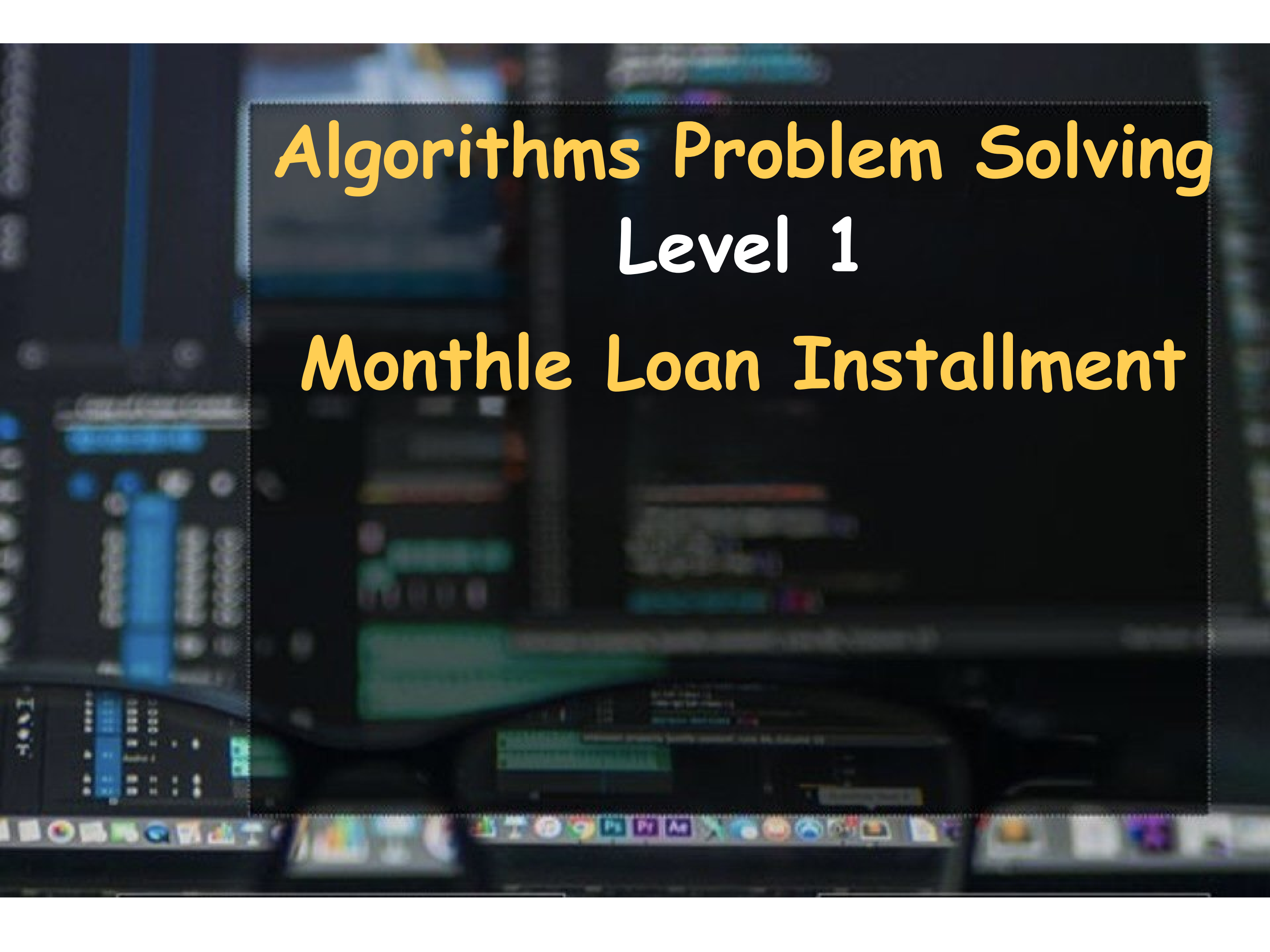




Algorithms Problem Solving Level 1

Monthle Loan Installment



Algorithms Problem Solving

Level 1

Monthle Loan Installment

Definition

◆ This program reads a loan amount and the number of months available to settle the loan, then calculates the required monthly payment.

◆ The calculation is:

$\text{LoanAmount} \div \text{NumberOfMonths}$

◆ The program uses a structure to group the loan information and separate functions for input, calculation, and output.

Explanation

- ◆ ``LoanInfo`` is a structure that stores:
 - `LoanAmount`
 - `Months`
- ◆ ``ReadInputs()`` asks the user for the loan amount and the number of months, then returns the information as a ``LoanInfo`` object.
- ◆ ``CalculateMonthlyPayment()`` divides the loan amount by the number of months.
- ◆ ``PrintResult()`` displays the calculated monthly payment.
- ◆ In ``main()``, the values from the structure are passed to the calculation function, and the returned result is passed to ``PrintResult()``.

Steps in main()

1. Create the `LoanInfo` structure.
2. Read the loan amount.
3. Read the number of months.
4. Return the loan information.
5. Divide the loan amount by the number of months.
6. Return the monthly payment.
7. Print the result.

💡 Example

◆ If the user enters:

Loan Amount: 5000

Months: 10

◆ The calculation is:

$$5000 \div 10 = 500$$

➡ Result

You will need to pay 500 per month to settle the loan.

Note

- ◆ The program assumes that the loan is divided equally across all months.
- ◆ Since the variables are `double`, the monthly payment can contain decimal values.

⊘ Warning

- ◆ The program does not validate the number of months. If `Months` is `0`, division by zero occurs.
- ◆ A negative number of months is also not meaningful for this calculation.



Comparison

Loan Amount

- ✓ Represents the total amount that needs to be paid.
- ✓ Example: 5000.
- ✓ It is divided by the number of months.

Monthly Payment

- ✓ Represents the amount paid each month.
- ✓ Example: 500 when 5000 is divided over 10 months.
- ✓ It is calculated rather than entered directly.

✦ Summary

- ✓ The monthly payment is calculated by dividing the loan amount by the number of months.
- ✓ A structure can group related loan information.
- ✓ Functions separate input, calculation, and output.
- ✓ The number of months must be greater than zero.

Thank you 😊